Jaewon Kim

Undergraduate Student, Korea University

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RESEARCH INTEREST

- Deep Reinforcement Learning
- Generative Models
- Physics-Informed Machine Learning
- AI for Science

EDUCATION

Korea University **Mar 2022 – Expected Aug 2028**

B.S. Semiconductor Engineering *Seoul, Korea*

- **GPA**: 4.35/4.5 (3.93/4.0)
- Tuition and stipend fully covered by SK Hynix-KU Partnership Fellowship.
- Military leave of absence for 2 academic years (Mar 2025 – Dec 2026).

University of Texas at Austin **Jan 2027 – Jun 2027**

B.S. Mathematics *Austin, Texas*

- Selected for UT Austin Student Exchange Program.

RESEARCH EXPERIENCE

Complex Data Analytics Lab (CDAL) **Feb 2024 – Present**

Undergraduate Researcher (Advisor: Prof. Seok Junhee) *Seoul, Korea*

- **High-Dimensional Conditional Generation.** Led the theoretical formulation and algorithmic design of the Hutchinson-Yielded Prioritized Estimator for Continuous Conditional Diffusion Models (`HYPE-CCDM` [**P1**]), generalizing continuous conditional generation from scalar to multi-dimensional regression labels. Overcame high-dimensional distance concentration by reformulating the vicinal objective with random 1D projections and distance-aware prioritized sampling, achieving state-of-the-art performance in both synthetic and real-world datasets.
- **Physics-Informed Policy Learning.** Proposed Fokker-Planck Soft Actor-Critic (`FP-SAC` [**J1**]), reinterpreting soft policy improvement as steering a policy toward the stationary solution of a Langevin SDE. Mitigated mode collapse in reverse-KL updates by training normalizing-flow policies via Fokker-Planck PDE residual minimization, enabling highly expressive multi-modal policies that significantly outperform traditional Gaussian-constrained approaches.

PUBLICATIONS

(P: Preprint, J: Journal, C: Conference, W: Workshop & Demos, *: Equal contribution, †: Equal advising)

[**P2**] **NanoDiff: Physics-Guided Diffusion for Image-Based Nanophotonic Device Design**
Insoo Kim, Jaewon Kim, Jangwon Seo, Junhee Seok†, Bohyung Han†
Under Review

[**P1**] **Scalable Continuous Conditional Diffusion via Random Projections and Prioritized Sampling**
Jaewon Kim*, Hyo-Seok Hwang*, Junhee Seok
Under Review

[**J1**] **Fokker-Planck Soft Actor-Critic**
Hyo-Seok Hwang, Jaewon Kim, Junhee Seok
IEEE Transactions on Neural Networks and Learning Systems, 2026

HONORS & AWARDS

SK Hynix-KU Partnership Fellowship Feb 2022 – Present

- Awarded full tuition and stipend through the selective SK Hynix-KU Fellowship.

KU Alumni Scholarship, Korea University Feb 2022 – Present

- Merit-based scholarship awarded to high-achieving students by the KU Alumni Association.

Semester High Honors, Korea University Mar 2022 – Present

- Awarded for achieving top academic performance each semester.

Outstanding Poster Award, 12th SK Hynix Academic Conference 2024

- Recognized for excellence in research presentation among selected poster submissions.

Dean's List, Korea University 2022

- Granted for achieving a perfect GPA (4.5/4.5) in the 1st semester of 2022.

EXTRACURRICULAR ACTIVITIES

Young Engineers' Honor Society (YEHS) Mar 2024 – Present

- Honor Society sponsored by the National Academy of Engineering of Korea (NAEK).

BITAmin Aug 2023 – Aug 2024

- Intercollegiate offline study group focusing on ML/DL research and projects

Korea University International Student Assistant (KUISA) Feb 2023 – Feb 2024

- Voluntary student organization under Global Services Center, Korea University.

SKILLS & INTERESTS

Language Skills: Korean(Native), English(Fluent)
Computer Skills: Python (Pytorch, Tensorflow), Matlab, C, C++
Interests: Music, Basketball, Badminton, Cooking, Traveling, Math

SELECTED COURSEWORKS

Statistics & Mathematics: Engineering Mathematics I, Engineering Mathematics II, Probability and Random Process, Introduction to Linear Algebra, Fundamentals of Data Science, Data Learning and Intelligence

Electrical and Computer Engineering: Introduction to Computers, Object-Oriented Programming, Microprocessor, Computer System, System Software, Signal and Systems, Digital Image Processing, Digital Signal Processing, Deep Signal Processing